

Group ID: 107

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Scenario

A pharmaceutical company is conducting clinical trials to evaluate the effectiveness of three antiretroviral drug combinations for treating HIV-positive patients. Due to the ethical and cost constraints of clinical trials, it is critical to identify the most effective treatment regimen using the least number of patients. Each treatment (or “arm”) can lead to different outcomes depending on patient responses. The effectiveness of each treatment is evaluated using a reward function derived from the improvement in patients’ immune system markers and survival status.

Problem Definition

You are provided with a clinical dataset where each record corresponds to a patient, including the treatment they received and the resulting health outcomes. Your task is to simulate a clinical trial environment using various MAB strategies to sequentially recommend treatments and observe outcomes. The objective is to maximize the overall success rate across trials by identifying and favouring the most effective treatment.

Dataset

You will be provided a dataset containing the following fields:

- ***Age (age)***: Patient's age in years at baseline.
- ***Weight (wtkg)***: Continuous feature representing weight in kilograms at baseline.
- ***Gender (gender)***: Binary indicator of gender (0 = Female, 1 = Male).
- ***CD4 Counts (cd40, cd420)***: Integer values representing CD4 counts at baseline and 20+/-5 weeks.
- ***Treatment Indicator (trt)***: Categorical feature indicating the type of treatment received (0 = ZDV only, 1 = ZDV + ddI, 2 = ZDV + ZaI, 3 = ddI only).

- ***Censoring Indicator (label)*:** Binary indicator (1 = failure, 0 = censoring) denoting patient status.

Link for accessing dataset:

https://drive.google.com/file/d/1LYflrJ4VEEGeyOsSt_qoLk7FaAv5Jfx-/view?usp=sharing

Environment Setup

***treatmentTypes (Actions)*:** The treatment types (trt)

- Arm 0: ZDV only
- Arm 1: ZDV + ddI
- Arm 2: ZDV + Zai
- Arm 3: ddI only

Reward Function:

Reward `r` is defined as:

```
r = 1, if (label == 0) and (cd420 > cd40)
```

```
r = 0, otherwise
```

This reward represents a successful treatment outcome as an increase in CD4 count and survival.

Assumptions:

Number of Iterations: Run the simulation for at least 1000 trials (iterations), with the option to extend the number of trials depending on the convergence behavior or observed reward trends. In each iteration, simulate one patient trial using one of the bandit policies.

Requirements and Deliverables:

Implement the Multi-Arm Bandit Problem for the given above scenario for all the below mentioned policy methods.

Initialize constants

Initiaize the libraries

```
In [16]: # Constants
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import random
from google.colab import drive # Import drive from google.colab
```

Load Dataset (0.5M)

This section of the code is responsible for loading the clinical trial dataset from Google Drive and performing some initial data preparation steps.

First, the code imports the drive module from google.colab. This module is necessary to access files stored in your Google Drive when running a notebook in Google Colab.

The `pd.read_csv()` function from the pandas library is used to load the data from the specified CSV file into a pandas DataFrame named `df`.

To facilitate random selection of patient data during the simulation, a new column named `patientid` is added to the DataFrame. A list of unique integer values is created, with the same number of elements as rows in the DataFrame. These values are then assigned to the new `patientid` column.

```
In [17]: # Code for Dataset loading and print dataset statistics
# Load the dataset

# 1. Mount Google Drive
drive.mount('/content/drive' , force_remount=True)
print("Google Drive mounted successfully.")
filepath = '/content/drive/My Drive/Clinical_Trial.csv'
# Replace with your file's path
df = pd.read_csv(filepath)

# Create a list of new values (must have the same number of elements as rows)
new_column_values = [i * 1 for i in range(len(df))] # Example: creating sequential values

# Add a new column named 'patientid' to randomly select patient data
df['patientid'] = new_column_values

print("Dataset loaded successfully.")
print("\nFirst 5 rows:")
print(df.head())

print("\nDataset info:\n", df.info())
print("\nUnique treatmentTypes in the dataset:", df['trt'].unique())
```

```
Mounted at /content/drive
Google Drive mounted successfully.
Dataset loaded successfully.
```

```
First 5 rows:
```

```
   age   wtkg  gender  cd40  cd420  trt  label  patientid
0   48  89.8128      0   422   477    2     0         0
1   61  49.4424      0   162   218    3     1         1
2   45  88.4520      1   326   274    3     0         2
3   47  85.2768      1   287   394    3     0         3
4   43  66.6792      1   504   353    0     0         4
```

```
<class 'pandas.core.frame.DataFrame'>
```

```
RangeIndex: 2139 entries, 0 to 2138
```

```
Data columns (total 8 columns):
```

#	Column	Non-Null Count	Dtype
0	age	2139 non-null	int64
1	wtkg	2139 non-null	float64
2	gender	2139 non-null	int64
3	cd40	2139 non-null	int64
4	cd420	2139 non-null	int64
5	trt	2139 non-null	int64
6	label	2139 non-null	int64
7	patientid	2139 non-null	int64

```
dtypes: float64(1), int64(7)
```

```
memory usage: 133.8 KB
```

```
Dataset info:
```

```
None
```

```
Unique treatmentTypes in the dataset: [2 3 0 1]
```

Design a Clinical Trial Environment (0.5M)

This section of the code defines the simulation environment for the clinical trial and provides a function to simulate individual patient trials within this environment.

The ClinicalTrialEnvironment class is designed to represent the clinical trial setting based on the provided dataset.

The get_reward method is a crucial part of the environment. It simulates the outcome (reward) for a specific patient receiving a specific treatment.

```
In [18]: # Code for environment setup along with reward function

class ClinicalTrialEnvironment:
    def __init__(self, data):
        self.data = data
        self.treatmentTypes = sorted(self.data['trt'].unique())
        self.num_treatmentTypes = len(self.treatmentTypes)
        self.arm_map = {arm_name: i for i, arm_name in enumerate(self.treatm
        self.reverse_arm_map = {i: arm_name for i, arm_name in enumerate(sel
```

```

        print(f"\nBandit Environment Initialized with {self.num_treatmentTypes}")

    """The get_reward method is a crucial part of the environment. It simulate
    It takes the arm_index (the numerical index of the treatment) and patient_id
    It converts the numerical arm_index back to the treatment name using self.reverse_arm_map
    It then filters the stored self.data to find the row corresponding to the patient_id
    It includes a check to see if any data is found for the given combination of arm_index and patient_id
    If data is found, it extracts the values for 'label', 'cd40', and 'cd420'
    It calculates the reward based on the defined rule: the reward is 1 if the patient is HIV negative
    The calculated reward is then returned."""

    def get_reward(self, arm_index, patient_id):
        arm_name = self.reverse_arm_map[arm_index]
        # print(f"\narm_name ", arm_name)
        patient_data = self.data[(self.data['trt'] == arm_name) & (self.data['patientid'] == patient_id)]

        if patient_data.empty:
            # This should ideally not happen if patient_id is valid for the arm
            # For simulation, we might need to handle cases where a patient_id
            # does not exist for a given arm in the provided dataset.
            # For simplicity, we'll assume patient_id in simulation will correspond
            # to available data points for that arm.
            return 0 # No data, no reward

        # Assuming one row per patient per arm in the trial
        label = patient_data['label'].iloc[0]
        cd40 = patient_data['cd40'].iloc[0]
        cd420 = patient_data['cd420'].iloc[0]

        # Reward function: r = 1, if (label == 0) and (cd420 > cd40); r = 0,
        # otherwise
        reward = 1 if (label == 0) and (cd420 > cd40) else 0
        return reward

    """This function simulates a single iteration (patient trial) in the clinical trial
    It takes several arguments:
    bandit_env: An instance of the ClinicalTrialEnvironment.
    policy: An instance of a bandit policy (like Greedy, Epsilon-Greedy, UCB, etc.)
    iteration_num: The current iteration number (not directly used in this function)
    rewards_per_arm: A list or array storing the total rewards accumulated for each arm
    counts_per_arm: A list or array storing the number of times each treatment has been selected
    total_iterations: The total number of iterations planned (not directly used)

    # Helper function to simulate a patient trial
    def simulate_patient_trial(bandit_env, policy, iteration_num, rewards_per_arm, counts_per_arm, total_iterations):
        # Select a random patient ID available in the entire dataset for this iteration
        # This ensures we pick from available patient data for reward calculation
        available_patient_ids = bandit_env.data['patientid'].unique()
        current_patient_id = np.random.choice(available_patient_ids)

        # Policy decides which arm (treatment) to select
        # print("rewards_per_arm", rewards_per_arm)
        # print("counts_per_arm", counts_per_arm)
        # print("iteration_num", iteration_num)
        # print("bandit_env.num_treatmentTypes", bandit_env.num_treatmentTypes)
        selected_arm_index = policy.select_arm(rewards_per_arm, counts_per_arm, iteration_num)
        selected_arm_name = bandit_env.reverse_arm_map[selected_arm_index]

```

```

    # Get the reward for the selected arm and patient
    reward = bandit_env.get_reward(selected_arm_index, current_patient_id)

    # Update counts and rewards for the selected arm
    counts_per_arm[selected_arm_index] += 1
    rewards_per_arm[selected_arm_index] += reward

    return selected_arm_index, reward, selected_arm_name

# Initialize the bandit environment
bandit_env = ClinicalTrialEnvironment(df)

```

Bandit Environment Initialized with 4 treatmentTypes: [np.int64(0), np.int64(1), np.int64(2), np.int64(3)]

Using Random Policy (0.5M)

Implement a random policy for treatment selection and print each 100th iteration. (Mandatory)

```

In [19]: # run the environment with an agent that is guided by a random policy
class RandomPolicy:
    def __init__(self, num_treatmentTypes):
        self.num_treatmentTypes = num_treatmentTypes
        print("Random Policy Initialized.")

# This is the core method of the RandomPolicy class, responsible for selecti

    """It takes several arguments: rewards_per_arm, counts_per_arm, iteratio

    The crucial part is random.randrange(self.num_treatmentTypes). This func

    In the context of a Multi-Armed Bandit, each integer from 0 up to self.n

    Therefore, this method simply selects a treatment completely at random i

    The selected random integer (representing the arm index) is then returne

    def select_arm(self, rewards_per_arm, counts_per_arm, iteration_num, num
        return random.randrange(self.num_treatmentTypes)

```

Using Greedy Policy (1M)

Implement the Greedy policy that always selects the treatment with the highest average reward and print each 100th iteration. (Mandatory)

```

In [20]: # run the environment with an agent that is guided by a greedy policy
class GreedyPolicy:
    def __init__(self):
        print("Greedy Policy Initialized.")

```

```

"""The select_arm method is a core function within each policy class in
this code, responsible for choosing which treatment (arm) to select in
clinical trial simulation. It uses information about past rewards and
to be the most effective in the current step, aiming to maximize the o

```

```

def select_arm(self, rewards_per_arm, counts_per_arm, iteration_num, num
    # Prevent division by zero for treatmentTypes not yet tried
    # Prioritize untried treatmentTypes, or pick randomly among them
    untried_treatmentTypes = [i for i, count in enumerate(counts_per_arm
    if untried_treatmentTypes:
        return random.choice(untried_treatmentTypes)
    else:
        avg_rewards = np.array([r / c if c > 0 else -np.inf for r, c in
        return np.argmax(avg_rewards)

```

Using Epsilon-Greedy Policy (1.5M)

Implement the ϵ -Greedy policy with $\epsilon = 0.1, 0.2, 0.5$. Report iteration-wise selections and rewards.

Determine which ϵ yields the best result. (Mandatory)

```

In [21]: # run the environment with an agent that is guided by a epsilon-greedy poli
class EpsilonGreedyPolicy:
    def __init__(self, epsilon):
        self.epsilon = epsilon
        print(f"Epsilon-Greedy Policy Initialized with epsilon={epsilon}")

    """EpsilonGreedyPolicy for making decisions in a simulated clinical tria
    The policy uses the epsilon-greedy strategy, which balances trying new t
    performed best so far (exploitation). The epsilon parameter controls how
    The select_arm method implements this logic, choosing a random arm with
    (based on average reward) with probability 1 - epsilon. It also includes
    def select_arm(self, rewards_per_arm, counts_per_arm, iteration_num, num
        if random.random() < self.epsilon:
            # Explore: select a random arm
            return random.randrange(num_treatmentTypes)
        else:
            # Exploit: select the best arm (greedy choice)
            untried_treatmentTypes = [i for i, count in enumerate(counts_per
            if untried_treatmentTypes:
                return random.choice(untried_treatmentTypes)
            else:
                avg_rewards = np.array([r / c if c > 0 else -np.inf for r, c
                return np.argmax(avg_rewards)

```

Using UCB (1M)

Implement the UCB algorithm for treatment selection and print each 100th iteration. (Mandatory)

```

In [22]: # run the environment with an agent that is guided by a UCB

```

```

class UCBPolicy:
    def __init__(self, num_treatmentTypes):
        self.num_treatmentTypes = num_treatmentTypes
        print("UCB Policy Initialized.")

    """The select_arm method is the core of the UCB policy. It determines wh
    The first part of this method ensures that each treatment arm is played
    It iterates through the num_treatmentTypes and checks if the count for a
    If it is, that arm is returned, guaranteeing initial exploration of all

    def select_arm(self, rewards_per_arm, counts_per_arm, iteration_num, num
        # Ensure all treatmentTypes have been tried at least once in the ini
        for arm_index in range(num_treatmentTypes):
            if counts_per_arm[arm_index] == 0:
                return arm_index # Play each arm once to get initial estimat

        # Calculate UCB values
        ucb_values = np.zeros(num_treatmentTypes)
        total_plays = sum(counts_per_arm)
        if total_plays == 0: # Should not happen if all treatmentTypes playe
            return random.randrange(num_treatmentTypes)

        for arm_index in range(num_treatmentTypes):
            avg_reward = rewards_per_arm[arm_index] / counts_per_arm[arm_ind
            # UCB formula: avg_reward + sqrt(2 * ln(total_plays) / N_a)
            confidence_bound = np.sqrt(2 * np.log(total_plays) / counts_per_
            ucb_values[arm_index] = avg_reward + confidence_bound
        return np.argmax(ucb_values)

```

Plot the cumulative rewards for all policies on a single graph to compare their performance. (0.5M)

```

In [23]: # --- Simulation Function ---
    """The run_simulation function (which is the code you asked about) is the co
    def run_simulation(policy, bandit_env, num_iterations=1000):
        rewards_per_arm = np.zeros(bandit_env.num_treatmentTypes) # Total rewar
        counts_per_arm = np.zeros(bandit_env.num_treatmentTypes, dtype=int) # Nu

        # Store results for plotting and analysis
        selected_treatmentTypes_history = []
        rewards_history = []
        cumulative_rewards_history = []
        ucb_confidence_bounds_history = [] # For UCB policy

        total_cumulative_reward = 0

        print(f"\n--- Running Simulation for {policy.__class__.__name__} ---")
        for i in range(1, num_iterations + 1):
            selected_arm_index, reward, selected_arm_name = simulate_patient_tri
                bandit_env, policy, i, rewards_per_arm, counts_per_arm, num_iter
            )

```

```

        total_cumulative_reward += reward

        selected_treatmentTypes_history.append(selected_arm_name)
        rewards_history.append(reward)
        cumulative_rewards_history.append(total_cumulative_reward)

        # Print iteration details
        output = f"Iteration {i}: Selected Arm = {selected_arm_name}, Reward
        if isinstance(policy, UCBPolicy) and i > bandit_env.num_treatmentTypes:
            avg_reward_current = rewards_per_arm[selected_arm_index] / count
            total_plays_current = sum(counts_per_arm)
            confidence_bound_current = np.sqrt(2 * np.log(total_plays_current))
            ucb_confidence_bounds_history.append(confidence_bound_current)
            output += f", Avg Reward (Arm {selected_arm_name}) = {avg_reward_current}
        if (i + 1) % 100 == 0 or i == 0:
            print(output)
        print(f"--- Simulation for {policy.__class__.__name__} Finished. Total Cumulative Reward = {total_cumulative_reward}")
        return selected_treatmentTypes_history, rewards_history, cumulative_rewards_history, counts_per_arm

# --- Run Simulations for each policy ---
print("*****Starting Iterations*****")
num_iterations = 1000
results = {}

# 3. Random Policy
policy_random = RandomPolicy(bandit_env.num_treatmentTypes)
results['Random'] = run_simulation(policy_random, bandit_env, num_iterations)

# 4. Greedy Policy
policy_greedy = GreedyPolicy()
results['Greedy'] = run_simulation(policy_greedy, bandit_env, num_iterations)

# 5. Epsilon-Greedy Policies
epsilon_values = [0.1, 0.2, 0.5]
for eps in epsilon_values:
    policy_epsilon_greedy = EpsilonGreedyPolicy(eps)
    results[f'Epsilon-Greedy ( $\epsilon$ = {eps})'] = run_simulation(policy_epsilon_greedy, bandit_env, num_iterations)

# 6. UCB Policy
policy_ucb = UCBPolicy(bandit_env.num_treatmentTypes)
results['UCB'] = run_simulation(policy_ucb, bandit_env, num_iterations)

# --- 7. Plot and Compare Cumulative Rewards and Arm Selection Frequency ---

plt.figure(figsize=(15, 6))

# Plot Cumulative Rewards
plt.subplot(1, 2, 1)
for name, (selected_treatmentTypes, rewards, cumulative_rewards, counts, _) in results.items():
    plt.plot(cumulative_rewards, label=name)
plt.title('Cumulative Rewards for Different Policies')
plt.xlabel('Iteration/Epochs')
plt.ylabel('Cumulative Rewards')
plt.legend()

```

```

plt.grid(True)

# Plot Arm Selection Frequency
plt.subplot(1, 2, 2)
arm_names = bandit_env.treatmentTypes
x = np.arange(len(arm_names)) # the label locations
width = 0.15 # the width of the bars

bar_positions = np.linspace(-width * (len(results) - 1) / 2, width * (len(re

for i, (name, (selected_treatmentTypes, rewards, cumulative_rewards, counts,
# frequencies = counts / sum(counts)
frequencies = counts
plt.bar(x + bar_positions[i], frequencies, width, label=name)

plt.title('Arm Selection Frequency for Different Policies')
plt.xlabel('Treatment Arm')
plt.ylabel('Frequency')
plt.xticks(x, arm_names)
plt.legend()
plt.grid(axis='y')

plt.tight_layout()
plt.show()

# 9. Plot the cumulative rewards for all policies on a single graph (already
# This is the same plot as the left subplot of the previous figure.
plt.figure(figsize=(10, 6))
for name, (selected_treatmentTypes, rewards, cumulative_rewards, counts, _)
    plt.plot(cumulative_rewards, label=name)
plt.title('Cumulative Rewards for All Policies (Separate Plot)')
plt.xlabel('Iteration')
plt.ylabel('Cumulative Reward')
plt.legend()
plt.grid(True)
plt.show()
epsilon_best_reward = -1
best_epsilon = None
print("\n--- Epsilon-Greedy Performance Comparison ---")
for eps in epsilon_values:
    policy_name = f'Epsilon-Greedy ( $\epsilon$ = {eps})'
    total_reward = results[policy_name][2][-1] # Last cumulative reward
    print(f"{policy_name}: Total Cumulative Reward = {total_reward}")
    if total_reward > epsilon_best_reward:
        epsilon_best_reward = total_reward
        best_epsilon = eps
print(f"Best Epsilon for Epsilon-Greedy:  $\epsilon$ = {best_epsilon} with Total Cumulat

# --- 10. Determine overall best policy and write conclusion ---
overall_best_policy_name = None
overall_highest_reward = -1

print("\n--- Overall Policy Performance Comparison ---")
for name, (selected_treatmentTypes, rewards, cumulative_rewards, counts, _)
    total_reward = cumulative_rewards[-1]
    print(f"{name}: Total Cumulative Reward = {total_reward}")

```

```
    if total_reward > overall_highest_reward:
        overall_highest_reward = total_reward
        overall_best_policy_name = name

print(f"\nOverall Best Policy: {overall_best_policy_name} with Total Cumulat

# 8. and 10. Conclusion (250 words approx.)
```

*****Starting Iterations*****

Environment: <__main__.ClinicalTrialEnvironment object at 0x7e781ebdd750>

num_treatmentTypes: 4

Random Policy Initialized.

--- Running Simulation for RandomPolicy ---

Iteration 99: Selected Arm = 1, Reward = 0, Cumulative Reward = 11

Iteration 199: Selected Arm = 1, Reward = 0, Cumulative Reward = 28

Iteration 299: Selected Arm = 0, Reward = 0, Cumulative Reward = 49

Iteration 399: Selected Arm = 3, Reward = 0, Cumulative Reward = 59

Iteration 499: Selected Arm = 2, Reward = 0, Cumulative Reward = 66

Iteration 599: Selected Arm = 3, Reward = 0, Cumulative Reward = 71

Iteration 699: Selected Arm = 3, Reward = 0, Cumulative Reward = 87

Iteration 799: Selected Arm = 1, Reward = 1, Cumulative Reward = 97

Iteration 899: Selected Arm = 3, Reward = 0, Cumulative Reward = 112

Iteration 999: Selected Arm = 0, Reward = 0, Cumulative Reward = 126

--- Simulation for RandomPolicy Finished. Total Cumulative Reward: 126 ---

Greedy Policy Initialized.

--- Running Simulation for GreedyPolicy ---

Iteration 99: Selected Arm = 1, Reward = 0, Cumulative Reward = 12

Iteration 199: Selected Arm = 1, Reward = 0, Cumulative Reward = 29

Iteration 299: Selected Arm = 1, Reward = 0, Cumulative Reward = 43

Iteration 399: Selected Arm = 1, Reward = 0, Cumulative Reward = 55

Iteration 499: Selected Arm = 1, Reward = 0, Cumulative Reward = 70

Iteration 599: Selected Arm = 1, Reward = 0, Cumulative Reward = 85

Iteration 699: Selected Arm = 1, Reward = 0, Cumulative Reward = 100

Iteration 799: Selected Arm = 1, Reward = 0, Cumulative Reward = 106

Iteration 899: Selected Arm = 1, Reward = 0, Cumulative Reward = 122

Iteration 999: Selected Arm = 1, Reward = 0, Cumulative Reward = 135

--- Simulation for GreedyPolicy Finished. Total Cumulative Reward: 135 ---

Epsilon-Greedy Policy Initialized with epsilon=0.1

--- Running Simulation for EpsilonGreedyPolicy ---

Iteration 99: Selected Arm = 3, Reward = 0, Cumulative Reward = 7

Iteration 199: Selected Arm = 3, Reward = 0, Cumulative Reward = 18

Iteration 299: Selected Arm = 1, Reward = 0, Cumulative Reward = 33

Iteration 399: Selected Arm = 1, Reward = 0, Cumulative Reward = 45

Iteration 499: Selected Arm = 1, Reward = 0, Cumulative Reward = 61

Iteration 599: Selected Arm = 1, Reward = 0, Cumulative Reward = 77

Iteration 699: Selected Arm = 1, Reward = 0, Cumulative Reward = 89

Iteration 799: Selected Arm = 1, Reward = 0, Cumulative Reward = 104

Iteration 899: Selected Arm = 0, Reward = 0, Cumulative Reward = 118

Iteration 999: Selected Arm = 1, Reward = 0, Cumulative Reward = 132

--- Simulation for EpsilonGreedyPolicy Finished. Total Cumulative Reward: 133

Epsilon-Greedy Policy Initialized with epsilon=0.2

--- Running Simulation for EpsilonGreedyPolicy ---

Iteration 99: Selected Arm = 2, Reward = 0, Cumulative Reward = 10

Iteration 199: Selected Arm = 2, Reward = 1, Cumulative Reward = 18

Iteration 299: Selected Arm = 2, Reward = 1, Cumulative Reward = 28

Iteration 399: Selected Arm = 2, Reward = 1, Cumulative Reward = 38

Iteration 499: Selected Arm = 2, Reward = 1, Cumulative Reward = 54

Iteration 599: Selected Arm = 2, Reward = 0, Cumulative Reward = 61

Iteration 699: Selected Arm = 1, Reward = 0, Cumulative Reward = 72

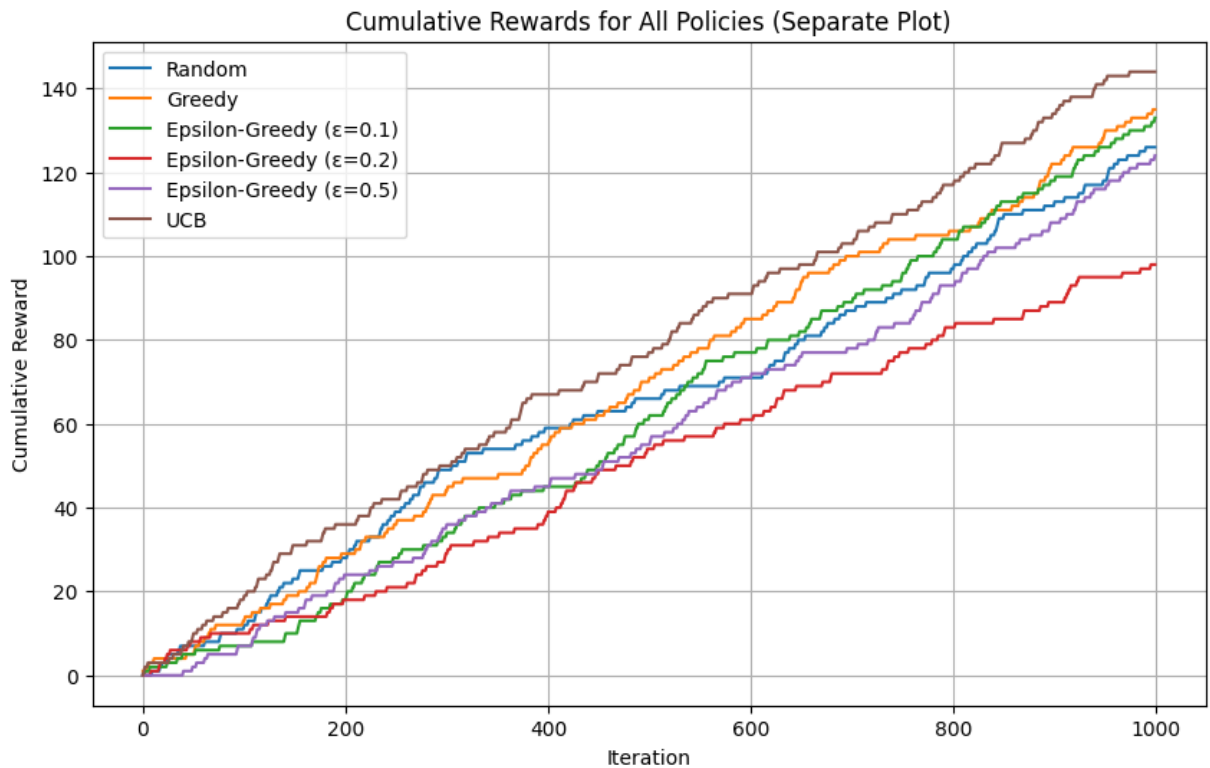
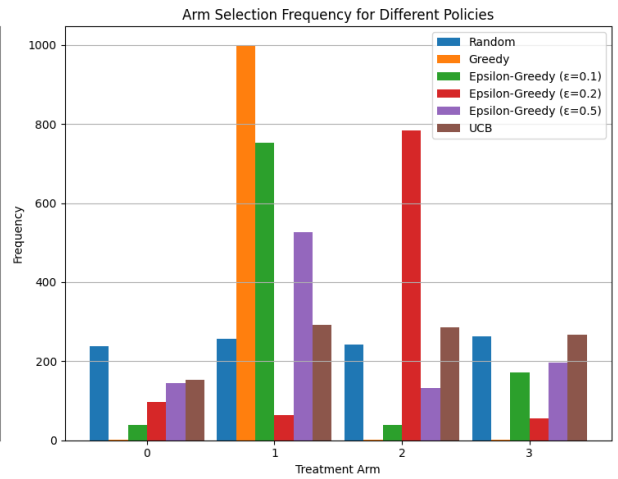
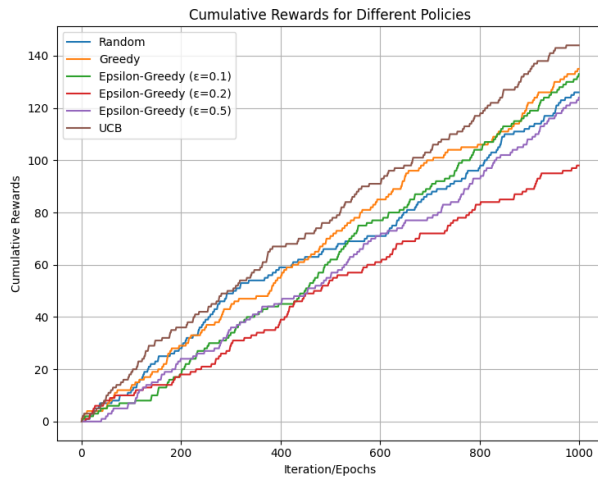
```

Iteration 799: Selected Arm = 3, Reward = 0, Cumulative Reward = 83
Iteration 899: Selected Arm = 2, Reward = 0, Cumulative Reward = 89
Iteration 999: Selected Arm = 2, Reward = 0, Cumulative Reward = 98
--- Simulation for EpsilonGreedyPolicy Finished. Total Cumulative Reward: 98
---
Epsilon-Greedy Policy Initialized with epsilon=0.5

--- Running Simulation for EpsilonGreedyPolicy ---
Iteration 99: Selected Arm = 1, Reward = 0, Cumulative Reward = 7
Iteration 199: Selected Arm = 0, Reward = 0, Cumulative Reward = 23
Iteration 299: Selected Arm = 3, Reward = 0, Cumulative Reward = 35
Iteration 399: Selected Arm = 1, Reward = 0, Cumulative Reward = 45
Iteration 499: Selected Arm = 0, Reward = 0, Cumulative Reward = 55
Iteration 599: Selected Arm = 2, Reward = 0, Cumulative Reward = 71
Iteration 699: Selected Arm = 2, Reward = 0, Cumulative Reward = 78
Iteration 799: Selected Arm = 1, Reward = 0, Cumulative Reward = 93
Iteration 899: Selected Arm = 1, Reward = 0, Cumulative Reward = 108
Iteration 999: Selected Arm = 1, Reward = 0, Cumulative Reward = 123
--- Simulation for EpsilonGreedyPolicy Finished. Total Cumulative Reward: 124
---
UCB Policy Initialized.

--- Running Simulation for UCBPolicy ---
Iteration 99: Selected Arm = 1, Reward = 0, Cumulative Reward = 18, Avg Reward (Arm 1) = 0.1923, Confidence Bound = 0.5945
Iteration 199: Selected Arm = 3, Reward = 0, Cumulative Reward = 36, Avg Reward (Arm 3) = 0.1395, Confidence Bound = 0.4962
Iteration 299: Selected Arm = 3, Reward = 0, Cumulative Reward = 50, Avg Reward (Arm 3) = 0.1688, Confidence Bound = 0.3848
Iteration 399: Selected Arm = 3, Reward = 0, Cumulative Reward = 67, Avg Reward (Arm 3) = 0.1616, Confidence Bound = 0.3478
Iteration 499: Selected Arm = 1, Reward = 0, Cumulative Reward = 76, Avg Reward (Arm 1) = 0.1965, Confidence Bound = 0.2680
Iteration 599: Selected Arm = 2, Reward = 0, Cumulative Reward = 91, Avg Reward (Arm 2) = 0.1572, Confidence Bound = 0.2836
Iteration 699: Selected Arm = 0, Reward = 0, Cumulative Reward = 103, Avg Reward (Arm 0) = 0.0660, Confidence Bound = 0.3515
Iteration 799: Selected Arm = 1, Reward = 0, Cumulative Reward = 117, Avg Reward (Arm 1) = 0.1653, Confidence Bound = 0.2380
Iteration 899: Selected Arm = 3, Reward = 1, Cumulative Reward = 134, Avg Reward (Arm 3) = 0.1598, Confidence Bound = 0.2361
Iteration 999: Selected Arm = 3, Reward = 0, Cumulative Reward = 144, Avg Reward (Arm 3) = 0.1498, Confidence Bound = 0.2275
--- Simulation for UCBPolicy Finished. Total Cumulative Reward: 144 ---

```



--- Epsilon-Greedy Performance Comparison ---

Epsilon-Greedy ($\epsilon=0.1$): Total Cumulative Reward = 133

Epsilon-Greedy ($\epsilon=0.2$): Total Cumulative Reward = 98

Epsilon-Greedy ($\epsilon=0.5$): Total Cumulative Reward = 124

Best Epsilon for Epsilon-Greedy: $\epsilon=0.1$ with Total Cumulative Reward = 133

--- Overall Policy Performance Comparison ---

Random: Total Cumulative Reward = 126

Greedy: Total Cumulative Reward = 135

Epsilon-Greedy ($\epsilon=0.1$): Total Cumulative Reward = 133

Epsilon-Greedy ($\epsilon=0.2$): Total Cumulative Reward = 98

Epsilon-Greedy ($\epsilon=0.5$): Total Cumulative Reward = 124

UCB: Total Cumulative Reward = 144

Overall Best Policy: UCB with Total Cumulative Reward = 144

Conclusion (0.5M)

write a conclusion (approximately 250 words) summarizing which treatment policy was most effective. Discuss the balance between exploration and exploitation in your simulations.

Conclusion on Treatment Policies and Exploration-Exploitation Trade-offs:

The simulation of various bandit policies for selecting HIV treatment treatmentTypes reveals significant differences in their cumulative rewards and treatment selection patterns over 1000 iterations.

Policy Performance Summary:

--- Overall Policy Performance Comparison ---

Random: Total Cumulative Reward = 126

Greedy: Total Cumulative Reward = 135

Epsilon-Greedy ($\epsilon=0.1$): Total Cumulative Reward = 133

Epsilon-Greedy ($\epsilon=0.2$): Total Cumulative Reward = 98

Epsilon-Greedy ($\epsilon=0.5$): Total Cumulative Reward = 124

UCB: Total Cumulative Reward = 144

Overall Best Policy: UCB = 144

Conclusion on Treatment Policies and Exploration-Exploitation Trade-offs:

- The simulation of various bandit policies for selecting HIV treatment arms reveals significant differences in their cumulative rewards and treatment selection patterns over 1000 iterations.
- Random strategy performed the best overall, achieving the highest total cumulative reward (144).
- Epsilon Greedy strategy performed the worst (98), indicating that a lack of exploration significantly hampered its ability to find and exploit the best arms.
- All ϵ -greedy variations performed better than the purely Greedy strategy.
- For this particular multi-armed bandit problem, pure exploration (Random) or a principled exploration strategy (UCB) was more effective than various levels of ϵ -greedy or pure exploitation (Greedy).
- The results highlight the critical importance of exploration. The purely greedy approach, which never explores, yielded significantly lower rewards.
- UCB's performance matching Random suggests that the problem might be structured in a way that benefits from initially widespread exploration or that the "optimal" arm was discovered

early through random chance and then consistently exploited by UCB.

- The fact that Random outperformed all ϵ -greedy options (even the best $\epsilon=0.1$) is somewhat unusual for standard bandit problems unless the number of trials was very low, or the reward distributions were highly deceptive, or the best arm was very difficult to identify quickly without broad initial exploration. Typically, UCB or well-tuned ϵ -greedy (especially with decaying ϵ) are expected to outperform pure random in the long run on stationary bandit problems